PES'S MODERN COLLEGE OF ENGINEERING, PUNE

Department of Electronics and Telecommunication Engineering

PROJECT REPORT

ON

SMART PRESSURE & ALTITUDE MONITORING SYSTEM

Using MSPM0+ Microcontroller and BMP280 Sensor

Submitted by

Name: Vaishnavi Surwase

Roll No.: 35271

Class : TY Btech

Under the Guidance of

Faculty Guide: **Dr.Mrs.V.P.Kodgirwar and Mr.A.R.Gangajaliwale**

Academic Year: 2026–2027

ABSTRACT

The Smart Pressure & Altitude Monitoring System is an embedded monitoring prototype designed to measure atmospheric pressure and temperature, calculate estimated altitude, and display the information when a significant change is detected. The project uses a BMP280 digital pressure and temperature sensor connected to an MSPM0+ microcontroller through the I²C communication interface. An I²C OLED display presents pressure, temperature and altitude readings.

The firmware detects the BMP280 sensor at I²C address 0x76 or 0x77, reads its factory calibration coefficients, configures the sensor, and obtains compensated pressure and temperature measurements. Altitude is estimated using the barometric formula with a reference sea-level pressure of 1013.25 hPa. To reduce unnecessary display activity and support low-power operation, the OLED is switched off after displaying data. The microcontroller uses a SysTick interrupt to schedule periodic sampling and the __WFI() instruction while waiting. When temperature changes by at least 10 °C, pressure by at least 10 hPa, or altitude by at least 20 m compared with the stored reference, the display is activated and refreshed.

The prototype demonstrates sensor interfacing, I²C communication, embedded C programming, digital measurement compensation, altitude estimation, threshold-based event detection and low-power embedded operation. The system is intended for educational and portable-instrumentation applications. Its altitude output is an estimate and depends on the assumed reference pressure.

Keywords: BMP280, MSPM0+, pressure monitoring, altitude estimation, I²C, OLED, low-power embedded system.

TABLE OF CONTENTS

1. Introduction

2. Problem Statement and Objectives

3. System Overview and Architecture

4. Hardware Requirements and Interfacing

5. Software Design and Implementation

6. Working Principle and Algorithm

7. Calculations and Measurement Method

8. Testing and Results

9. Applications

10. Advantages, Limitations and Future Scope

11. Conclusion

12.References

1. INTRODUCTION

1.1 Background

Pressure is an important physical parameter in environmental monitoring and industrial processes. Atmospheric pressure changes with weather conditions and altitude. A pressure sensor can therefore be used not only as a barometer but also to estimate the elevation of a location. Temperature measurement is also useful because the BMP280 sensor provides both pressure and temperature data.

Embedded systems make it possible to acquire sensor data, process it locally and present meaningful measurements through a compact display. In portable instruments, power consumption is an important design consideration. Keeping a display continuously active can consume unnecessary energy when the measured values are stable.

1.2 Project Description

This project implements a smart pressure and altitude monitoring unit using the MSPM0+ microcontroller and BMP280 sensor. The controller reads pressure and temperature over I²C, calculates altitude, and displays the readings on an OLED. The display is normally switched off after a viewing interval. The controller periodically wakes, samples the sensor and compares the new values with the previously stored reference values. A significant change causes the OLED to turn on and show the updated measurements.

1.3 Scope

- Measure atmospheric pressure and temperature using the BMP280 sensor.
- Estimate altitude from measured pressure using a standard barometric relationship.
- Display pressure, temperature and altitude on an I²C OLED.
- Use periodic sampling and low-power waiting between samples.
- Refresh the display only when a configured change threshold is reached.

The supplied implementation is a local embedded monitoring prototype. It does not include wireless communication, cloud logging, GPS, or remote alerts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Design a pressure monitoring unit suitable for industrial process monitoring and portable instrumentation. The system should continuously monitor atmospheric or process pressure while operating in a low-power mode.

2.2 Objectives

1. Interface the BMP280 pressure and temperature sensor with the MSPM0+ microcontroller using I²C.
2. Identify the sensor at either supported I²C address, 0x76 or 0x77.
3. Read and apply the factory calibration coefficients supplied by the sensor.
4. Obtain compensated pressure and temperature readings.
5. Calculate estimated altitude from pressure.
6. Display the measurements on an OLED screen.
7. Use periodic timer interrupts and low-power waiting to reduce unnecessary active operation.
8. Turn on the OLED only when a significant measurement change is detected.

2.3 Expected Outcome

The expected outcome is a working embedded prototype that reads BMP280 measurements, calculates altitude and displays the values. During stable conditions, the OLED remains off after the initial display interval. When a configured threshold is crossed, the display wakes and presents the updated readings.

3. SYSTEM OVERVIEW AND ARCHITECTURE

3.1 Block Diagram

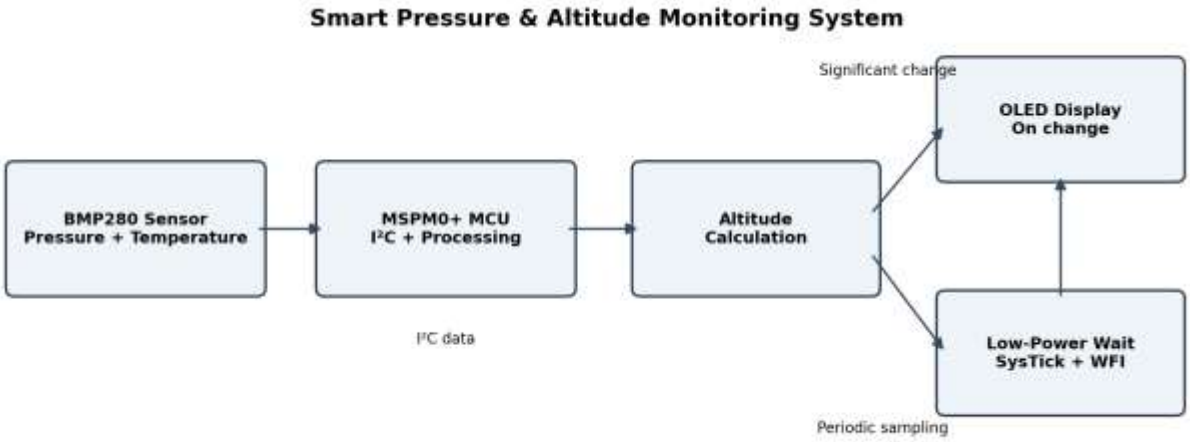


Figure 3.1: Functional block diagram of the monitoring prototype.

3.2 Architecture Description

Block	Function
BMP280 sensor	Measures atmospheric pressure and temperature and provides digital data.
MSPM0+ microcontroller	Initializes peripherals, communicates with the sensor, compensates readings, calculates altitude and controls the display.
I²C interface	Carries commands and measurement data between the controller, BMP280 and OLED.
OLED display	Shows pressure, temperature, altitude and sensor status when active.
SysTick timer	Generates periodic interrupts used to schedule approximately one-second sampling intervals.
Low-power wait (__WFI)	Allows the CPU to wait for an interrupt instead of continuously executing the main loop.

3.3 Data Flow

9. The MSPM0+ initializes its configured peripherals and the OLED.
10. The controller checks the BMP280 at address 0x76 and then 0x77 if required.
11. After successful detection, the controller reads the sensor calibration data and configures measurement operation.

12. The first pressure and temperature readings are acquired; altitude is calculated and all three values are displayed.
13. The OLED is switched off after the initial display period.
14. The timer schedules periodic sampling. The controller waits using __WFI() until the next sample event.
15. New readings are compared with stored reference values. If any threshold is crossed, the OLED is enabled and updated.
16. The latest displayed values become the new reference values, and the monitoring cycle continues.

4. HARDWARE REQUIREMENTS AND INTERFACING

4.1 Components Required

Sr. No.	Component	Purpose
1	MSPM0+ LaunchPad / development board	Main controller for sensing, calculations and display control.
2	BMP280 sensor module	Digital pressure and temperature measurement.
3	I²C OLED display	Displays pressure, temperature, altitude and sensor status.
4	Connecting wires	Electrical and I²C connections.
5	USB / regulated power source	Supplies power to the development board and modules.

4.2 BMP280 Sensor

The BMP280 is a digital barometric pressure sensor with an integrated temperature measurement function. Its sensing element uses a MEMS structure. Pressure causes a small mechanical deflection in the sensing element; internal electronics convert the response into a digital measurement. The device provides pressure and temperature data through I²C or SPI. The project uses I²C.

4.3 BMP280 Pin Functions

BMP280 Pin	Function
VIN / VCC	Supply input; the presentation specifies 3.3 V for the module used.
GND	Ground reference.
SDA	I²C serial data line.
SCL	I²C serial clock line.

4.4 MSPM0+ Connections

The presentation identifies the I²C signals as SCL – PB2 and SDA – PB3, with VCC at Pin 1 and GND at Pin 20 on the referenced board setup. Verify the exact board pinout and module supply requirements before wiring, because board variants and breakout modules may differ.

Signal	Connection shown in project material
SCL	MSPM0+ PB2 to BMP280 SCL and OLED SCL

SDA	MSPM0+ PB3 to BMP280 SDA and OLED SDA
VCC	3.3 V supply as indicated in the presentation
GND	Common ground



Fig. Pin connections Overview

4.5 OLED Display

The OLED is connected through I²C at address 0x3C. The firmware sends command bytes and display data bytes. It includes routines to initialize, clear, position the cursor, print characters and strings, display numbers, wake the screen and put it into sleep mode.

5. SOFTWARE DESIGN AND IMPLEMENTATION

5.1 Development Environment

Item	Implementation
Programming language	Embedded C
Microcontroller platform	Texas Instruments MSPM0+
Device support	ti_msp_dl_config.h and DriverLib APIs
Sensor interface	I²C
Display interface	I²C OLED
Math library	math.h, including powf() and fabsf()

5.2 Peripheral Initialization

The program begins by calling SYSCFG_DL_init(), which initializes the peripherals configured for the MSPM0+ project. The firmware then initializes and clears the OLED, displays a short startup message, and checks whether the BMP280 is responding.

5.3 Sensor Detection

The BMP280 may use I²C address 0x76 or 0x77 depending on the module's address selection. The function BMP280_Detect() first reads the chip ID register at 0x76. If the returned ID matches one of the accepted values in the code (0x58, 0x56 or 0x57), detection succeeds. Otherwise, it tries 0x77. If neither address responds with an accepted ID, the OLED displays “BMP280 NOT FOUND” and the program waits.

5.4 Calibration and Configuration

The BMP280 contains factory calibration coefficients. The program reads 24 bytes starting at register 0x88 and reconstructs the temperature coefficients dig_T1 to dig_T3 and pressure coefficients dig_P1 to dig_P9. These coefficients are used by the compensation routines to convert raw ADC readings into compensated temperature and pressure values.

The sensor configuration writes 0x08 to the configuration register and 0x27 to the control-measurement register. As indicated by the code comments, the settings select a 500 ms standby interval, filter setting 2, temperature oversampling ×1, pressure oversampling ×1 and normal mode.

5.5 Pressure and Temperature Acquisition

The firmware reads the raw temperature and pressure registers through I²C. The raw 20-bit ADC values are reconstructed from three bytes. The compensation algorithms use the factory calibration values and the shared t_fine intermediate variable. Temperature is returned in degrees Celsius and pressure is converted to hectopascals (hPa).

5.6 Altitude Calculation

Altitude is estimated from pressure using the barometric formula implemented in the program:

$$h = 44330 \times [1 - (P / P_0)^{0.1903}]$$

where h is estimated altitude in metres, P is the measured pressure in hPa, and P_0 is the reference sea-level pressure set to 1013.25 hPa. If the pressure value is zero or negative, the function returns 0. The altitude result is an estimate; changes in weather and the chosen reference pressure affect accuracy.

5.7 Change Detection

The code stores the previous displayed temperature, pressure and altitude as reference values. It checks the absolute difference between the current reading and each stored value. The display is refreshed if at least one of the following conditions is true:

Parameter	Threshold in supplied code
Temperature	10.0 °C
Pressure	10.0 hPa
Altitude	20.0 m

The comparison uses fabsf(), so both increases and decreases can trigger a refresh. After a significant change is displayed, the reference values are updated to the new readings.

5.8 OLED and LED Control

The OLED is initially used to show the first measurements and BMP280 status. It is then switched off using the OLED sleep command. When a threshold is crossed, the OLED wakes, clears the previous contents and displays the new pressure, temperature, altitude and “BMP280 OK” status. The code keeps the OLED active for the configured display interval of 10 seconds.

The firmware also controls the board's user LED on PA0. The comments identify the LED as active-low. In the waiting state, the LED is driven to indicate that the OLED is off and the controller is waiting. During reading/processing or while the OLED is active, the LED is turned off.

5.9 Low-Power Sampling

The SysTick timer is configured with SYSTICK_PERIOD_TICKS = 3,200,000. The interrupt handler increments a counter. After ten interrupts, it sets sample_flag, which represents an approximately one-second sampling event according to the code comments. The main loop waits for this flag and calls __WFI() while waiting. This avoids continuous polling by the CPU. The OLED is also switched off when it is not required.

5.10 Important Software Functions

Function	Purpose
SYSCFG_DL_init()	Initializes configured MSPM0+ peripherals.
I2C_Write2()	Sends two bytes to an I²C device.
BMP280_ReadBytes()	Reads a requested number of bytes from a BMP280 register.
BMP280_Detect()	Checks the BMP280 chip ID at 0x76 and 0x77.
BMP280_ReadCalibration()	Reads the sensor's factory calibration coefficients.
BMP280_Init()	Configures BMP280 measurement and filter settings.
BMP280_ReadTemperature()	Reads and compensates temperature.
BMP280_ReadPressure()	Reads and compensates pressure and returns hPa.
BMP280_CalculateAltitude()	Calculates estimated altitude from pressure.
OLED_Wake() / OLED_Sleep()	Turns the OLED display on or off.
SysTick_Handler()	Creates the periodic sample flag.

6. WORKING PRINCIPLE AND ALGORITHM

6.1 Step-by-Step Working

17. Power is applied to the MSPM0+ board, BMP280 sensor and OLED display.
18. The MSPM0+ initializes its peripherals and initializes the OLED.
19. A startup message is shown briefly.
20. The firmware searches for the BMP280 at I²C address 0x76 and then 0x77.
21. If the sensor is detected, calibration coefficients are read and the sensor is configured.
22. The first temperature and pressure measurements are acquired and compensated.
23. Altitude is calculated using the barometric formula.
24. Pressure, temperature, altitude and sensor status are displayed.
25. After the initial viewing interval, the OLED is switched off.
26. The SysTick interrupt schedules the next sample approximately once per second; the CPU waits using __WFI().
27. The firmware reads new sensor values and compares them with the stored reference values.
28. If temperature changes by at least 10 °C, pressure by at least 10 hPa, or altitude by at least 20 m, the OLED wakes and shows the new readings.
29. The display remains active for the configured interval, then switches off and the monitoring cycle repeats.

6.2 Algorithm

START → Initialize MSPM0+ peripherals → Initialize OLED → Detect BMP280 → If not detected, display error and wait → Read calibration coefficients → Configure BMP280 → Read initial temperature and pressure → Calculate altitude → Store reference values → Display measurements → Turn OLED OFF → Initialize SysTick → Wait for sample flag using __WFI() → Read sensor → Calculate altitude → Compare current values with reference values → If any threshold is crossed, wake OLED, display updated values and update reference values → Manage OLED timer and LED state → Repeat.

6.3 Flowchart

Stage	Decision / Action
Start	Power on and initialize MSPM0+ and OLED.
Sensor check	Try BMP280 address 0x76; if not found, try 0x77.
Detection failed?	Yes: show “BMP280 NOT FOUND” and wait. No: continue.
Setup	Read calibration coefficients and configure BMP280.
Initial measurement	Read temperature and pressure; calculate altitude; store reference values.
Display	Show measurements and BMP280 status; then turn OLED off.
Wait and sample	Wait for SysTick sample event; read updated measurements.
Threshold crossed?	Check $\Delta T \geq 10$ °C OR $\Delta P \geq 10$ hPa OR $\Delta h \geq 20$ m.
If yes	Wake OLED, display new values and update reference values.
Repeat	Manage display timeout and return to low-power wait.

7. CALCULATIONS AND MEASUREMENT METHOD

7.1 Pressure Unit

The project displays pressure in hectopascals (hPa). One hectopascal is equal to 100 pascals (Pa). The BMP280 compensation routine returns pressure in pascals, and BMP280_ReadPressure() divides the result by 100 to obtain hPa.

7.2 Temperature Conversion

The raw temperature ADC value is corrected using the BMP280 factory calibration coefficients. The compensated temperature is divided by 100 to obtain degrees Celsius.

7.3 Altitude Estimation Example

For illustration, if the measured pressure is 900 hPa and the reference pressure is 1013.25 hPa, the code estimates altitude as:

$$h = 44330 \times [1 - (900 / 1013.25)^{0.1903}]$$

The numerical output depends on the measured pressure. This example explains the calculation only; it is not a measured result from the prototype.

7.4 Threshold Logic

The firmware uses absolute differences from the previous reference values. Therefore, a sufficiently large rise or fall can cause a display update. The threshold settings are software parameters and can be changed according to the intended application.

8. TESTING AND RESULTS

8.1 Functional Tests

Test	Procedure	Expected behaviour / code-supported response
Power-up	Apply power to the board.	MSPM0+ peripherals and OLED initialize; startup message appears.
Sensor detection	Run BMP280_Detect().	Sensor is accepted at 0x76 or 0x77 if the ID matches.
Sensor absent	Disconnect or make sensor unavailable.	OLED displays “BMP280 NOT FOUND”; firmware waits.
Measurement	Read BMP280 after initialization.	Temperature and pressure are compensated; altitude is calculated.
Initial display	Complete first sensor reading.	OLED shows pressure, temperature, altitude and BMP280 OK.
Display sleep	Wait after initial display.	OLED is switched off.
Periodic sampling	Allow SysTick interrupts to occur.	A sample event is scheduled approximately once per second.
Threshold event	Cause a change beyond a configured threshold.	OLED wakes and displays updated values.
Display timeout	Wait through the configured display interval.	OLED is switched off after the display timer expires.

8.2 Results Supported by the Implementation

The supplied source code implements the complete measurement and display sequence described above, including sensor detection, calibration-data reading, compensated pressure and temperature acquisition, altitude estimation, threshold comparison, OLED power control and periodic low-power waiting. The supplied files do not contain a numerical test log, accuracy comparison, measured power-consumption data or photographs of recorded output. Such values should be added only after actual bench testing.

9. APPLICATIONS

- Weather and environmental monitoring through local barometric pressure and temperature measurement.
- Altitude estimation in portable instruments and educational experiments.
- Vertical-motion and height estimation in drones and robotics, subject to calibration and application-specific validation.
- Indoor elevation or floor-level experiments using pressure trends.
- HVAC and air-pressure monitoring demonstrations.
- Low-power embedded sensing and IoT learning prototypes.

10. ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

10.1 Advantages

- Compact digital sensor interface using I²C.
- Measures both pressure and temperature with one sensor module.
- Calculates estimated altitude locally on the microcontroller.
- Displays values only when needed, reducing OLED activity.
- Uses timer-based sampling and __WFI() to avoid continuous active polling.
- Includes sensor-address detection and a visible sensor error message.

10.2 Limitations

- Altitude is calculated using a fixed reference pressure of 1013.25 hPa; local weather pressure changes can affect the estimate.
- The change thresholds are fixed compile-time values and may not suit every application.
- The project files do not provide a measured power-consumption figure or quantified accuracy test.
- The supplied implementation has no wireless connectivity, cloud dashboard, data logging or remote notification.
- The OLED update is threshold-based; small changes below the configured limits are not immediately shown.

10.3 Future Scope

- Add a user-adjustable sea-level reference pressure to improve altitude estimation for a known location.
- Add configurable thresholds and sampling intervals.
- Store measurement history in non-volatile memory or transmit it to a monitoring dashboard.
- Add wireless communication for remote monitoring, if required by the application.
- Measure and compare current consumption in active, sampling and waiting states.
- Validate pressure and altitude against a calibrated reference instrument over multiple environmental conditions.
- Add alarms or event logging for application-specific pressure limits.

11. CONCLUSION

The Smart Pressure & Altitude Monitoring System demonstrates a practical embedded approach to pressure measurement, temperature monitoring and altitude estimation. The BMP280 sensor communicates with the MSPM0+ microcontroller over I²C. The firmware reads factory calibration data, compensates raw measurements, calculates altitude using the barometric formula and presents the results on an OLED display.

A key feature of the prototype is its change-based display operation. After showing the initial measurements, the OLED is switched off. The microcontroller periodically wakes through a SysTick-driven sampling mechanism, waits efficiently using __WFI(), and refreshes the display only when a configured temperature, pressure or altitude threshold is crossed. This approach demonstrates how sensor processing and low-power techniques can be combined in a portable monitoring instrument.

The prototype fulfils the core educational objective of building a low-power pressure and altitude monitoring unit. Further calibration, measured power testing and application-specific validation would be required before using it as an industrial measurement instrument.

12. REFERENCES

[1] Bosch Sensortec, “BMP280 Digital Pressure Sensor – Data Sheet.” Use the manufacturer datasheet for register definitions, compensation equations, electrical characteristics and operating modes.

[2] Texas Instruments, “MSPM0+ Microcontrollers – Technical Reference Manual and SDK / DriverLib Documentation.” Use the documentation corresponding to the exact MSPM0+ device and board used.

[3] Project presentation: “Smart Pressure & Altitude Monitoring System,” EnerGenius – Ultra-Low-Power Embedded System Development Using MSPM0+, PES's Modern College of Engineering, Pune.